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Automatic Quantity Takeoff for Irregular Areas of Architectural, Engineering, and Construction Objects from 2D Drawings

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Abstract

Accurate cost estimation of construction projects relies on precise quantity takeoffs (QTOs), which conventionally has been conducted on 2D drawings, and despite the fast development of Building Information Modeling (BIM), remains a predominant practice. This study focuses on the development of a boundary detection algorithm utilizing computer vision techniques for extracting the shapes of infrastructure elements such as buildings and bridges from 2D PDF drawings. The aim is to enhance information extraction capabilities, particularly for QTOs of complex shapes such as the areas of irregular bridge decks. Utilizing computer vision techniques, the algorithm aims to accurately delineate the boundaries of various infrastructure components by eliminating redundant lines, such as measurement lines and dashed lines. The proposed boundary detection algorithm achieved 94.1% accuracy in automatic area calculation, and about 99% time reduction compared to manual approach. The proposed approach holds promise for improving efficiency and accuracy in QTO processes, and offering potential solutions in more accurate QTOs.

Keywords

Computer Vision, Information Extraction, Quantity Takeoff

1 Introduction

Quantity takeoffs (QTOs) play a crucial role in cost estimation for architecture, engineering, and construction (AEC) projects (Alathamneh et al. 2024). Traditionally, QTOs are conducted manually on 2D drawings, which could be a time-consuming and error-prone process. Although Building Information Modeling (BIM) offers enhanced automation opportunities with improved interoperability by leveraging invariant signatures of AEC objects (Li et al. 2022a; Li et al. 2022b; Zhang and Yang 2024), its adoption varies across project types (e.g., buildings, bridges, drainage systems). There is a significant demand for automating information extraction and QTO in 2D bridge drawings (Alathamneh et al. 2024; Li and Zhang 2024). Challenges persist in extracting accurate quantities, particularly for irregular and complex shapes in 2D drawings.

The extraction of irregular deck areas is based on the federal requirement: Specifications for the National Bridge Inventory (SNBI) (U.S. DOT 2022). SNBI provides standardized definitions and formats for bridge inventory items including span material, foundation type, and irregular deck area,

etc., facilitating reliable reporting to the Federal Highway Administration (FHWA) (U.S. DOT 2022). It ensures consistency, accuracy, and compliance with federal mandates for bridge data representation, supporting efficient bridge management and enhances safety (Li and Zhang 2024). The federal deadline of March 15, 2028, underscores the urgency of aligning with SNBI standards with the help of assistive technologies such as semantic information extraction from 2D bridge drawings (U.S. DOT 2023; Li and Zhang 2024).

In this research, the authors propose a novel algorithm leveraging machine learning-based and rule-based techniques to automate boundary detection and irregular deck area extraction from 2D bridge drawings. The approach focuses on optimizing the accuracy of QTOs and reducing the time required for processing complex shapes.

2 Literature Review

Automated techniques for extracting quantities and recognizing architectural elements from technical drawings and models have been extensively explored in previous research. For example, Xiao et al. (2020) leveraged Fully Convolutional Network (FCNs) in recognizing and segmenting architectural elements from CAD drawings, achieving an accuracy of 80%. Khosakitchalert et al. (2019) addressed the challenges of extracting accurate quantities from BIM for compound elements like walls and floors. They proposed the “BIM-based compound element quantity takeoff improvement” (BCEQTI) method, which utilizes BIM-based clash detection to automatically eliminate excessive quantities and add missing ones. Their method was validated through four case studies, demonstrating the effectiveness in providing accurate material quantities and saving time in model editing. The BCEQTI method is validated through four case studies, showing that the method significantly reduces deviations in material quantities compared to traditional and initial building information models (BIMs). In most cases, the deviation was reduced to less than $\pm 1\%$.

Akanbi et al. (2020) introduced the Data-Driven Reverse Engineering Algorithm Development (D-READ) method as a solution to overcome the limitations in QTO algorithm development for industry foundation classes (IFC)-based BIMs. The D-READ method adopts a bottom-up strategy, leveraging geometric analysis of IFC models to derive QTO algorithms. This approach enhances the robustness of QTO processes across BIMs created using diverse authoring tools and workflows. The integration of a Model View Definition (MVD) for IFC model checking further strengthens the method, enabling accurate extraction of building component quantities.

In another study by Akanbi and Zhang (2023), a case study was performed on a residential project in Kalamazoo, Michigan, which demonstrated that the developed IFC-based QTO algorithms were platform-agnostic, outperforming commercial software in interoperability while achieving similar results in extracting volumetric and areal quantities from various building configurations. These algorithms not only ensured consistency across different BIM platforms but also streamlined workflows by reducing manual input through direct integration with IFC data. The study underscores the potential of IFC-based algorithms to enhance the accuracy, automation, and scalability of QTO processes in construction projects.

While BIM-based and IFC-based methods offer robust solutions, their applicability is restricted to projects readily adopting BIM or IFC (Li and Zhang 2022). Manual QTO remains prevalent for 2D drawings especially for irregular shaped objects.

3 Methodology

The proposed method combines machine learning-based and rule-based techniques to automatically detect irregular deck areas in 2D bridge drawings. It includes the following key components as shown in Figure 1.

- (1) Machine Learning-Based Arrow Detection: a computer vision algorithm is trained to automatically detect the arrows in the drawing. The measurement lines (identified through associated arrows) are then removed from the drawing. This step ensures a cleaner drawing for more accurate identification of bridge deck edges.
- (2) Rule-Based Edge Detection and Contour Analysis: The irregular deck area boundaries are identified and isolated from redundant lines. Following that, the largest contour is determined to approximate the irregular deck area.

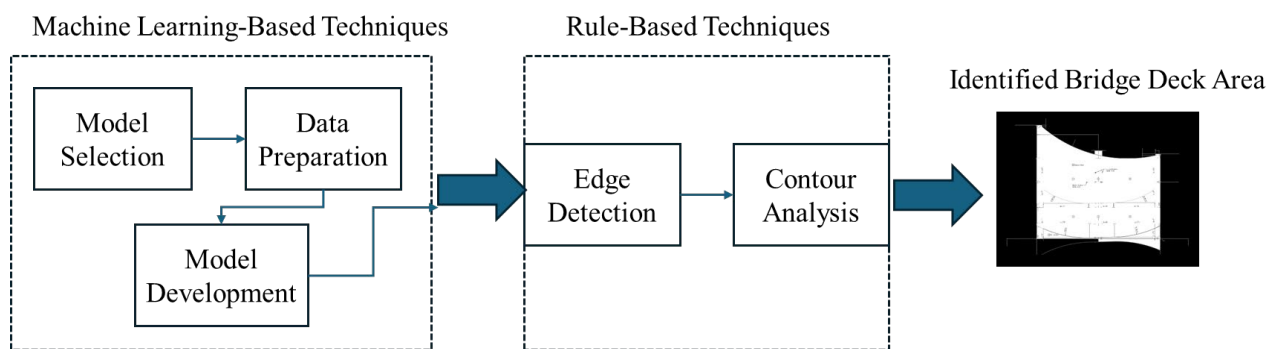


Figure 1. Proposed methodology for automatically detecting irregular deck area combining machine learning-based and rule-based techniques.

3.1 Machine learning-based Arrow Detection

3.1.1 Model Selection

YOLO (You Only Look Once) is a deep learning-based object detection algorithm famous for its speed and accuracy (Redmon et al. 2016). YOLO has advantages in processing the entire image in one pass and predicting bounding boxes and class probabilities simultaneously, compared with traditional methods that scan images in multiple steps (Redmon et al. 2016). This makes it efficient for real-time applications while maintaining competitive accuracy. YOLO has been widely used in various applications such as real-time autonomous vehicles detection (Alahdal et al. 2024) and damage detection in built structures (Raushan et al., 2024). In addition, its versatility and customizability allows it to be trained to identify specific arrow styles or orientations commonly used in bridge drawings. Therefore, YOLO model was selected in this study.

3.1.2 Data Preparation

A dataset comprising 2001 images was collected from the roboflow public datasets, with an 88%, 5%, and 7% split for training, validation, and testing, respectively (Roboflow 2024). Figure 2 (a) shows an example image with labelled arrows from the training dataset. Images were preprocessed by auto-orienting to ensure images are correctly aligned and oriented before training. This step automatically detected and corrected the orientation of an image, which could have been captured in different orientations (e.g., rotated, upside down, etc.).

Following that, the images were resized to 640x640 by pixel, because YOLO requires a consistent input size for all images. The model's architecture is designed to process images of fixed sizes, and

resizing all images to a common dimension ensures that the network receives input data in a uniform format. This allows the model to efficiently process images in batches during training and inference. It also helps the YOLO model generalize better, reduces training time, and is compatible with pre-trained YOLO models. When resizing, it's crucial to maintain the aspect ratio of the original image, either by padding or stretching. The aspect ratio was preserved by adding padding (black borders) to the shorter side, as shown in Figure 2 (b). Therefore, the objects' relative positions and sizes are maintained.

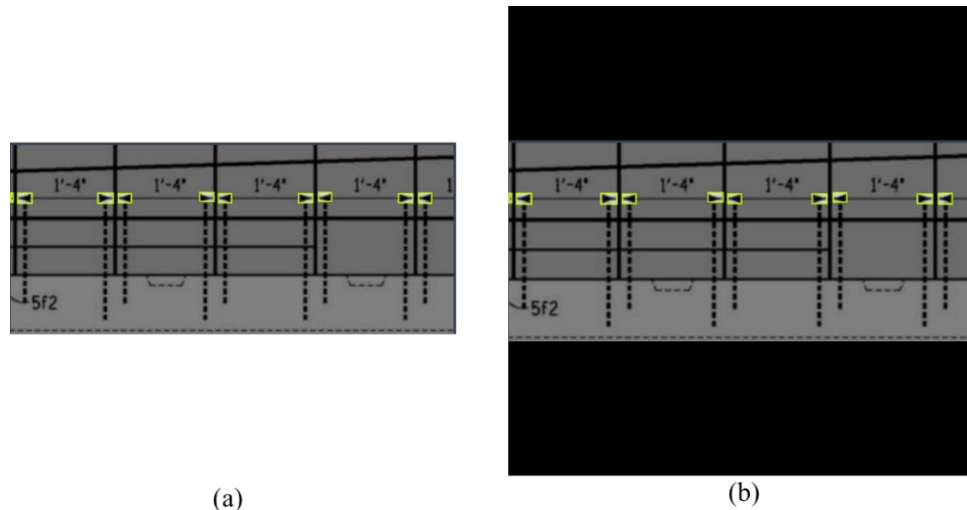


Figure 2. (a) Example original image with labelled arrows from the training dataset, and (b) resized 640x640 image with padding.

3.1.3 Model Development

The YOLO model was employed to train the machine learning-based algorithm using the collected data. It achieved precision and recall of 95.5% and 92.0%, respectively, for detecting arrows in the drawings. The trained YOLO model was then applied to the bridge deck plan to identify the arrows. Following that, measurement lines associated with the arrows were removed in the drawing.

3.2 Rule-based Edge Detection and Contour Analysis

3.2.1 Edge Detection

Canny edge detection was then applied for detecting the bridge deck area in this study. Canny edge detection is a powerful technique for identifying structural boundaries (OpenCV 2024a). By applying Canny edge detection, the gradient of the intensity at each pixel is computed to determine the strength and direction of these changes, as shown in Figure 3 (OpenCV 2024a). Gradients within the bridge plan can be analyzed to highlight edges that delineate the deck from other elements such as railings, supports, or the background.

Preprocessing steps including grayscale conversion and Gaussian blurring were applied to reduce noise and improve edge clarity. The result is a binary image including an approximate outline of the bridge deck, enabling further analysis such as segmentation, area measurement, and contour analysis.

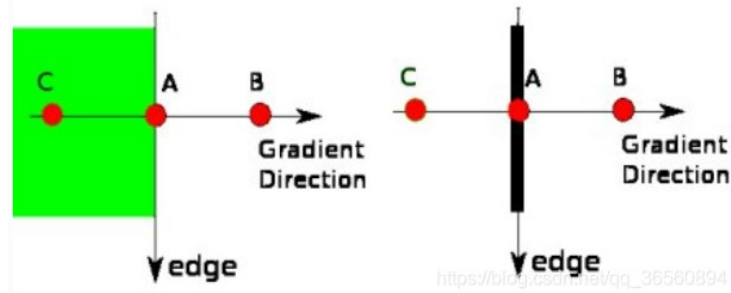


Figure 3. Illustration of Canny Edge Detection (OpenCV 2024a).

3.2.2 Contour Analysis

To identify the irregular deck area, contours are then extracted using a contour-finding algorithm using OpenCV (OpenCV 2024b). The area of each contour is computed and the contour with the maximum area is selected to represent the irregular deck area.

4 Case Study

The proposed methodology was tested on a bridge plan with an irregular deck area (126th Street over SR-37, northeast of Indianapolis), as shown in Figure 4.

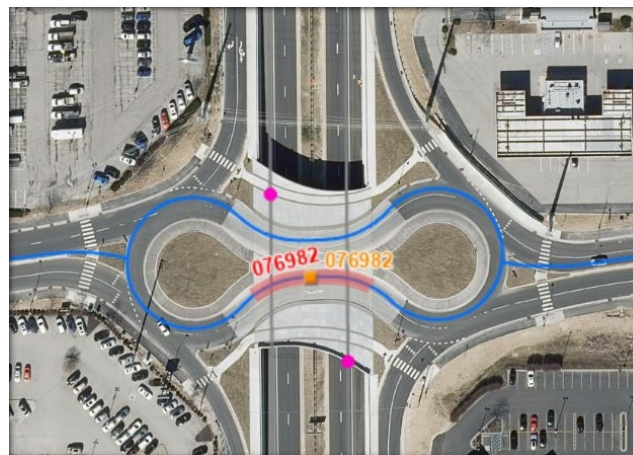


Figure 4. Example bridge for irregular deck area (126th Street over SR-37, northeast of Indianapolis, in Hamilton County).

Figure 5 (a) and (b) show the north bridge deck area and south bridge deck area, respectively. The irregular deck area boundary is highlighted in the blue bounding box.

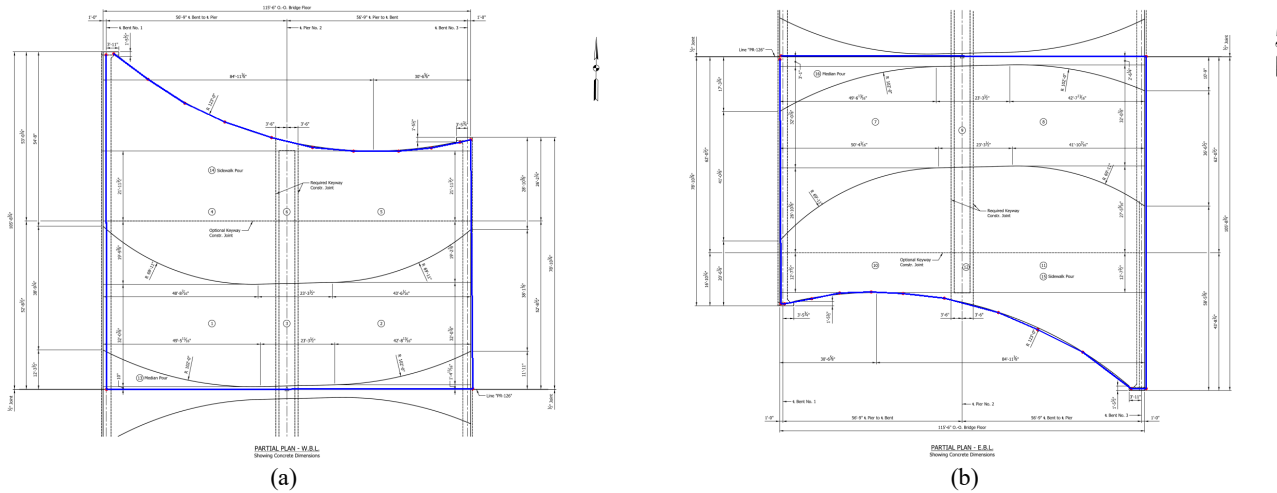


Figure 5. Detailed bridge plan with irregular deck area highlighted – (a) north bound and (b) south bound.

The original plan was then divided into 176 sub-images of size 640×640 as per the required input size for the trained YOLO model, as shown in Figure 6 (a). Figure 6 (b) shows the predicted arrows using the trained YOLO model. It can be seen that the arrows of the measurement lines were detected and labelled with the corresponding bounding box.

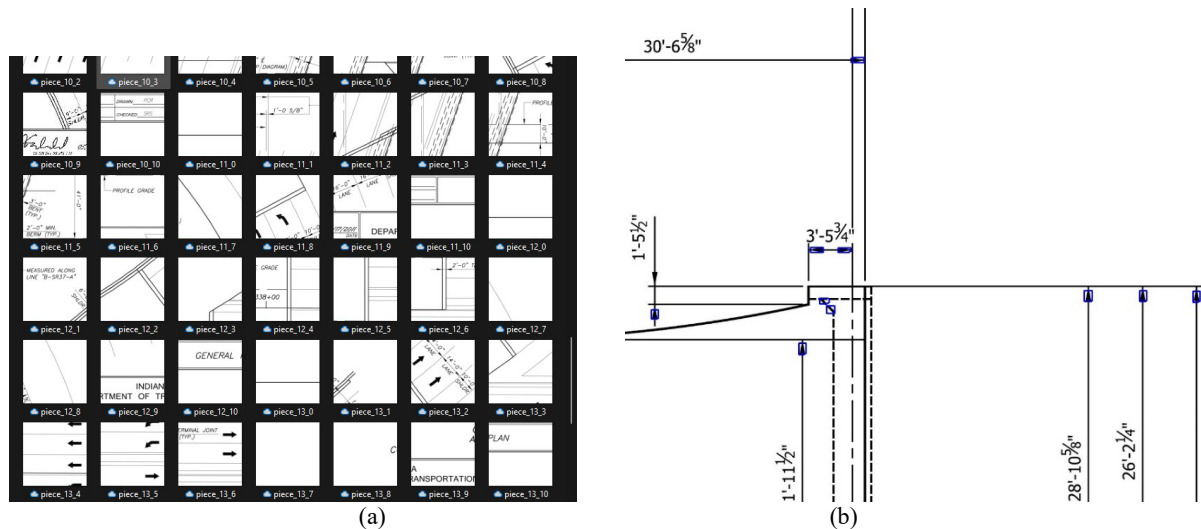


Figure 6. (a) Division of the original plan into 176 sub-images (640×640) and (b) detection of the arrows using the trained YOLO model (a part of the plan 640×640).

Following that, the measurement lines associated with the detected arrows were then removed from the drawings. Figure 7 (a) shows the intermediate results by applying arrow detection and edge detection and Figure 7 (b) shows the final detected area using the proposed algorithmic method.

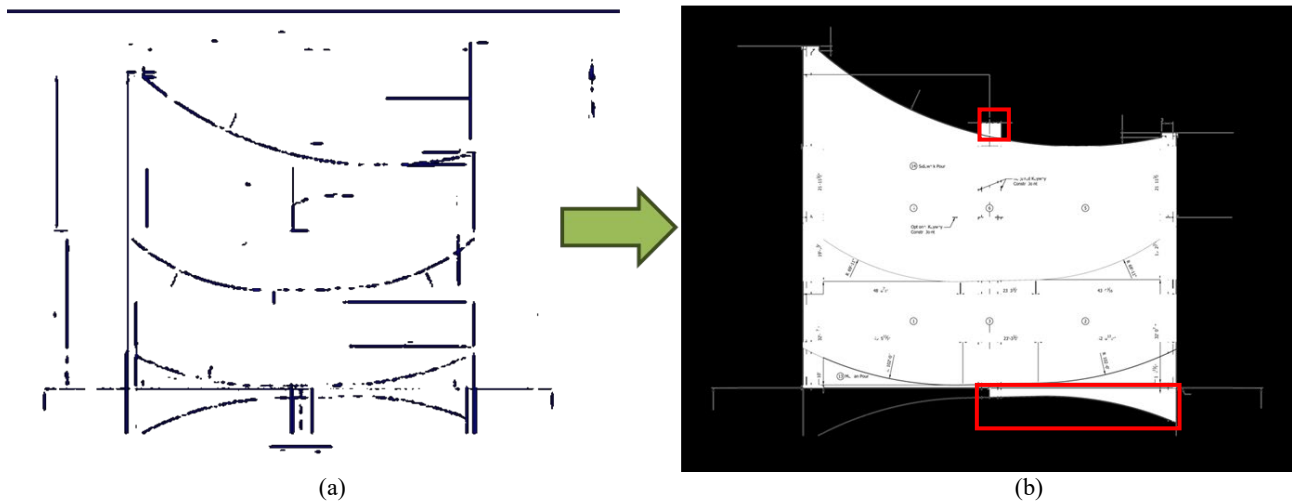


Figure 7. (a) intermediate results by applying edge detection and (b) final detected area (white) by applying contour analysis, using the proposed algorithmic method.

Manual takeoff was carried out using an onscreen takeoff software as the ground truth. Table 1 shows the comparison of time consumption and results for irregular deck area takeoff between manual approach and the proposed algorithmic approach. It can be seen that the time consumption has been reduced by 99.08%. In terms of results, the calculated area differed between the two methods. The manual method estimated the area as 10,190 ft², whereas the algorithmic method calculated it as 10,801 ft², with a 94.1% accuracy.

Table 1. Comparison of time consumption and QTO results for irregular deck area takeoff between manual approach and algorithmic approach.

	Manual method	Algorithmic method	Percentage difference
Time consumption	25 seconds	0.23 seconds	99.08%
QTO results (area)	10,190 ft ²	10,801 ft ²	94.1%

The error in the results obtained by the algorithmic method can be attributed to several factors that could have affected the overall accuracy of boundary detection and area calculation:

- (1) The arrow detection YOLO model has not achieved 100% precision and recall yet. Some arrows were missed by the algorithm. Redundant measurement lines can distort the identified contours of the bridge deck, introducing errors in area calculation (highlighted in the red boxes in Figure 7). Further improvements in the YOLO model, such as expanding the training dataset to include more diverse arrow styles and orientations, could enhance its robustness and reduce such errors.
- (2) The current algorithm identifies the largest contour as the deck boundary, assuming it represents the primary area of interest. This approach may fail in situations where multiple large contours are present, such as overlapping or adjacent elements. A more sophisticated contour selection strategy, incorporating contextual information or user-defined constraints, could help address this limitation.
- (3) The irregularity of bridge deck shapes presents additional challenges for the edge detection and contour analysis steps. Subtle variations in the deck's boundary, combined with closely spaced elements including railings, supports, or annotations, can lead to incorrect contour extraction. For example, small gaps or overlaps in the detected edges may cause the algorithm to either

overestimate or underestimate the boundary area. Future work can include implementing advanced noise-reduction techniques or adaptive thresholding to improve edge clarity and reduce such errors.

5 Conclusion

This study presents a novel approach for automating boundary detection and QTO processes for irregular deck area in bridge drawings. By leveraging a combination of machine learning-based arrow detection and rule-based edge detection and contour analysis, the proposed methodology addresses the challenges of accurately and efficiently extracting irregular bridge deck areas from 2D drawings. The case study on an irregular bridge deck area demonstrated the effectiveness of the proposed algorithm, achieving a 99.08% reduction in time consumption compared to the manual approach and delivering an accuracy of 94.1% in area calculation. These results underscore the algorithm's potential to streamline QTO processes, reducing reliance on labor-intensive and error-prone manual methods.

Despite its promising performance, the algorithm's limitations, including inaccuracies in arrow detection, challenges with contour selection, and the inherent complexity of irregular shapes, indicate opportunities for further refinement. Improvement to the YOLO model, more sophisticated contour analysis methods, and advanced edge detection techniques can further improve robustness and accuracy in the future work.

In conclusion, the proposed algorithm offers a promising solution for automating QTO processes in bridge projects. This work contributes to bridging the gap between traditional 2D drawing-based practices and modern, automated techniques, paving the way for broader adoption in the construction industry.

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7 References

- Akanbi, T., and Zhang, J. 2023. IFC-Based Algorithms for Automated Quantity Takeoff from Architectural Model: Case Study on Residential Development Project. *Journal of Architectural Engineering*, 29(4). <https://doi.org/10.1061/jaeied.aeeng-1447>
- Akanbi, T., Zhang, J., and Lee, Y. 2020. Data-Driven Reverse Engineering Algorithm Development Method for developing interoperable quantity takeoff algorithms using IFC-Based BIM. *Journal of Computing in Civil Engineering*, 34(5). [https://doi.org/10.1061/\(asce\)cp.1943-5487.0000909](https://doi.org/10.1061/(asce)cp.1943-5487.0000909)
- Alahdal, N. M., Abukhodair, F., Meftah, L. H., and Cherif, A. 2024. Real-time Object Detection in Autonomous Vehicles with YOLO. *Procedia Computer Science*, 246, 2792–2801. <https://doi.org/10.1016/j.procs.2024.09.392>

- Alathamneh, S., Collins, W., and Azhar, S. 2024. BIM-based quantity takeoff: Current state and future opportunities. *Automation in Construction*, 165, 105549. <https://doi.org/10.1016/j.autcon.2024.105549>
- Khosakitchalert, C., Yabuki, N., and Fukuda, T. 2019. Improving the accuracy of BIM-based quantity takeoff for compound elements. *Automation in Construction*, 106, 102891. <https://doi.org/10.1016/j.autcon.2019.102891>
- Li, H., J. Zhang, S. Chang, and A. Sparkling. 2022a. BIM-based object mapping using invariant signatures of AEC objects. *Automation in Construction*, 145, 104616. <https://doi.org/10.1016/j.autcon.2022.104616>.
- Li, H., J. Zhang, X. Xue, L. Debs, S. Chang, M. Qu, A. Sparkling, and D. Goldwasser. 2022b. Issues in Bi-Directional Interoperability between BIM and BEM. *Construction Research Congress 2022*, 1355–1364. <https://doi.org/10.1061/9780784483961.142>.
- Li, H., and J. Zhang. 2022. Interoperability between BIM and BEM Using IFC. *Computing in Civil Engineering*, 630–637. <https://doi.org/10.1061/9780784483893.078>.
- Li, H., and Zhang, J. 2024a. Information extraction for semantic enrichment of BIM for bridges. *Construction Research Congress 2022*, 629–638. <https://doi.org/10.1061/9780784485262.064>
- Li, H., and Zhang, J. 2024b. Semi-automatic IFC-based BIM Reconstruction for As-designed Bridges from 2D Plans Leveraging Semantic Segmentation and Enrichment. *Journal of Computing in Civil Engineering*, in press. DOI: 10.1061/JCCEE5/CPENG-5931.
- OpenCV: Canny Edge Detection. 2024a. Retrieved from https://docs.opencv.org/4.x/da/d22/tutorial_py_canny.html
- OpenCV: Contours : Getting started. 2024b. Retrieved from https://docs.opencv.org/3.4/d4/d73/tutorial_py_contours_begin.html.
- Roboflow. 2024. Computer Vision Datasets. Retrieved from <https://public.roboflow.com/>
- Redmon, J., Divvala, S., Girshick, R., and Farhadi, A. 2016. You Only Look Once: Unified, Real-Time Object Detection. *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 779–788). <https://doi.org/10.1109/cvpr.2016.91>
- Raushan, R., Singhal, V., and Jha, R. K. 2024. Damage detection in concrete structures with multi-feature backgrounds using the YOLO network family. *Automation in Construction*, 170, 105887. <https://doi.org/10.1016/j.autcon.2024.105887>
- U.S. Department of Transportation - Federal Highway Administration (U.S. DOT). 2022. Specifications for the national bridge inventory. *Office of Bridges and Structures*. https://www.fhwa.dot.gov/bridge/snbi/snbi_march_2022_publication.pdf
- U.S. Department of Transportation - Federal Highway Administration (U.S. DOT). 2023. Questions and Answers on the Specification for the National Bridge Inventory. *U.S. Department of Transportation/Federal Highway Administration*. <https://www.fhwa.dot.gov/bridge/snbi/qanda/02.cfm>
- Xiao, Y., S. Chen, Y. Ikeda, and K. Hotta. 2020. Automatic Recognition and Segmentation of Architectural Elements from 2D Drawings by Convolutional Neural Network. *Proceedings of the International Conference on Computer-Aided Architectural Design Research in Asia*. <https://doi.org/10.52842/conf.caadria.2020.1.843>.
- Zhang, J., and Yang, F. 2024. Building a Bridge between Building Information Modeling and Digital Twins: Introducing Invariant Signatures of Architecture, Engineering, and Construction Objects. *American Society of Civil Engineers eBooks*, 37–62, <https://doi.org/10.1061/9780784485606.ch2>